WORKED SOLUTION

STEP III 2010, Question 6

Pure • Rotations in Three Dimensions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q6

The key idea

A rotation about an axis leaves every point on the axis where it is, and turns everything in the plane perpendicular to the axis exactly as a rotation of the plane would. So each step reduces to a two-dimensional rotation: find the plane being turned, write down an orthonormal pair of vectors in it, and rotate them. Any vector along the axis is simply left alone, and that one observation does a lot of the work for Q.

Solution

Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ for the unit vectors along the axes, so P_0, Q_0, R_0 have position vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

Part (i)

Rotating about the z-axis turns the x-y plane through ϕ, with $\mathbf{i}$ moving towards $\mathbf{j}$. Hence

$$P_1 = (\cos\phi,\ \sin\phi,\ 0).$$

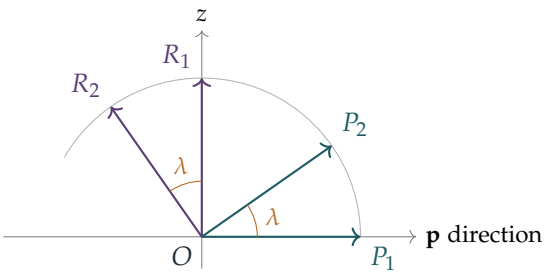
For later, the same rotation sends $\mathbf{j}$ to $Q_1 = (-\sin\phi, \cos\phi, 0)$, since Q stays a right angle ahead of P in that plane, and leaves R_0 fixed because it lies on the axis: $R_1 = (0, 0, 1)$.

Part (ii)

Let $\mathbf{p} = \overrightarrow{OP_1} = (\cos\phi, \sin\phi, 0)$. The line in the x-y plane perpendicular to OP_1 has direction

$$\mathbf{u} = (-\sin\phi,\ \cos\phi,\ 0),$$

since $\mathbf{u} \cdot \mathbf{p} = 0$ and $\mathbf{u} \cdot \mathbf{k} = 0$. The plane perpendicular to this axis is the plane spanned by the orthonormal pair $\mathbf{p}$ and $\mathbf{k}$, and the second rotation turns this plane through λ with $\mathbf{p}$ moving towards $\mathbf{k}$:



In that plane the rotation sends

$$\mathbf{p} \mapsto \cos\lambda\,\mathbf{p} + \sin\lambda\,\mathbf{k}, \qquad \mathbf{k} \mapsto -\sin\lambda\,\mathbf{p} + \cos\lambda\,\mathbf{k},$$

the second because $\mathbf{k}$ stays a right angle ahead of $\mathbf{p}$. Therefore

$$P_2 = (\cos\lambda\cos\phi,\ \cos\lambda\sin\phi,\ \sin\lambda), \qquad R_2 = (-\sin\lambda\cos\phi,\ -\sin\lambda\sin\phi,\ \cos\lambda).$$

Ask yourself Where does Q go in the second rotation?

After the first rotation Q is at $Q_1 = (-\sin\phi, \cos\phi, 0)$, which is exactly $\mathbf{u}$: it lies *on* the axis of the second rotation, so it does not move. Hence

$$Q_2 = (-\sin\phi,\ \cos\phi,\ 0).$$

As a check, P_2, Q_2, R_2 are mutually perpendicular unit vectors (for instance $\overrightarrow{OP_2} \cdot \overrightarrow{OR_2} = -\sin\lambda\cos\lambda\cos^2\phi - \sin\lambda\cos\lambda\sin^2\phi + \sin\lambda\cos\lambda = 0$), as they must be after rotations.

Part (iii)

Throughout this part we assume $P_2 \neq \pm P_0$, so that O, P_0, P_2 really do determine a plane.

Ask yourself Why is the angle of rotation the same as the angle P_2OP_0?

Since P stays in the plane OP_0P_2, the axis of the rotation must be the line through O perpendicular to that plane (a rotation moves a point in the plane perpendicular to the axis through that point). So P travels along the great circle in this plane, and every vector in the plane is turned through the angle of the rotation. That angle is therefore the angle between $\overrightarrow{OP_2}$ and $\overrightarrow{OP_0}$, and since both are unit vectors,

$$\cos\theta = \overrightarrow{OP_2} \cdot \overrightarrow{OP_0} = (\cos\lambda\cos\phi)(1) + (\cos\lambda\sin\phi)(0) + (\sin\lambda)(0) = \cos\phi\cos\lambda,$$

as required. (Turning the long way round, through $2\pi - \theta$, gives the same cosine.)

The axis is perpendicular to both $\overrightarrow{OP_2}$ and $\overrightarrow{OP_0}$, so it is along their cross product:

$$\overrightarrow{OP_2} \times \overrightarrow{OP_0} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\lambda\cos\phi & \cos\lambda\sin\phi & \sin\lambda \\ 1 & 0 & 0 \end{vmatrix} = (0,\ \sin\lambda,\ -\cos\lambda\sin\phi).$$

So a vector along the axis is

$$\begin{pmatrix} 0 \\ \sin\lambda \\ -\cos\lambda\sin\phi \end{pmatrix}.$$

With this orientation, a rotation through θ by the right-hand rule takes P_2 to P_0. Notice the axis has no x-component, as it must, since it is perpendicular to OP_0.

Checks

- If $\lambda = 0$ the second rotation does nothing, so the third just undoes the first: $\theta = \phi$ and the axis is along $(0, 0, -\sin\phi)$, the z-axis. The formulas agree.
- If $\phi = 0$ the only rotation was about the y-axis, and indeed the axis is along $(0, \sin\lambda, 0)$ and $\theta = \lambda$.
- Numerically, applying the rotations by Rodrigues' formula for 20 000 random (ϕ, λ) reproduces P_2, Q_2, R_2, and the rotation through θ about $(0, \sin\lambda, -\cos\lambda\sin\phi)$ returns P_2 to P_0. For $\phi = 0.7$, $\lambda = 0.4$, $\cos\theta \approx 0.7045$, so $\theta \approx 0.789$.

Common slips

- Rotating Q in the second step: it is on the axis, so it stays put.
- Getting the sign of R_2 wrong: as P tips up towards $\mathbf{k}$, R tips down and away from OP_1, so its horizontal part is $-\sin\lambda\,\mathbf{p}$.

Beyond STEP

The two angles here are longitude ϕ and latitude λ, and $\cos\theta = \cos\phi\cos\lambda$ is the right-angled case of the spherical cosine rule, used in navigation to find great-circle distances. More generally, the composition of two rotations is again a rotation about a single axis (Euler's rotation theorem), and the columns $\overrightarrow{OP_2}$, $\overrightarrow{OQ_2}$, $\overrightarrow{OR_2}$ form the orthogonal matrix of the combined rotation. Strang's *Introduction to Linear Algebra* treats rotation matrices and orthogonal matrices; Needham's *Visual Complex Analysis* shows how rotations of the sphere can be described with complex numbers.

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